

1.  $G(s) = \frac{100}{s^2 + 2s + 100}$

$|K G(j\omega)| = K \frac{100}{(100 - \omega^2)^2 + (2\omega)^2}$

b. Gain crossover:  $|K G(j\omega)| = 1$

If  $\text{PM} = 50^\circ$  then  $\phi(\omega_{\text{crossover}}) = -130^\circ + 50^\circ = -13^\circ$

$\angle (100 - \omega^2 + j2\omega)$

$= \tan^{-1} \left( \frac{2\omega}{100 - \omega^2} \right) = 130^\circ$

$\frac{2\omega}{100 - \omega^2} = \tan(130^\circ)$

$0 = -\omega^2 \tan(130^\circ) - 2\omega + 100 \tan(130^\circ)$

Solve by desmos:  $\omega = 10.878$  because  $\omega$  must be  $\geq 0$

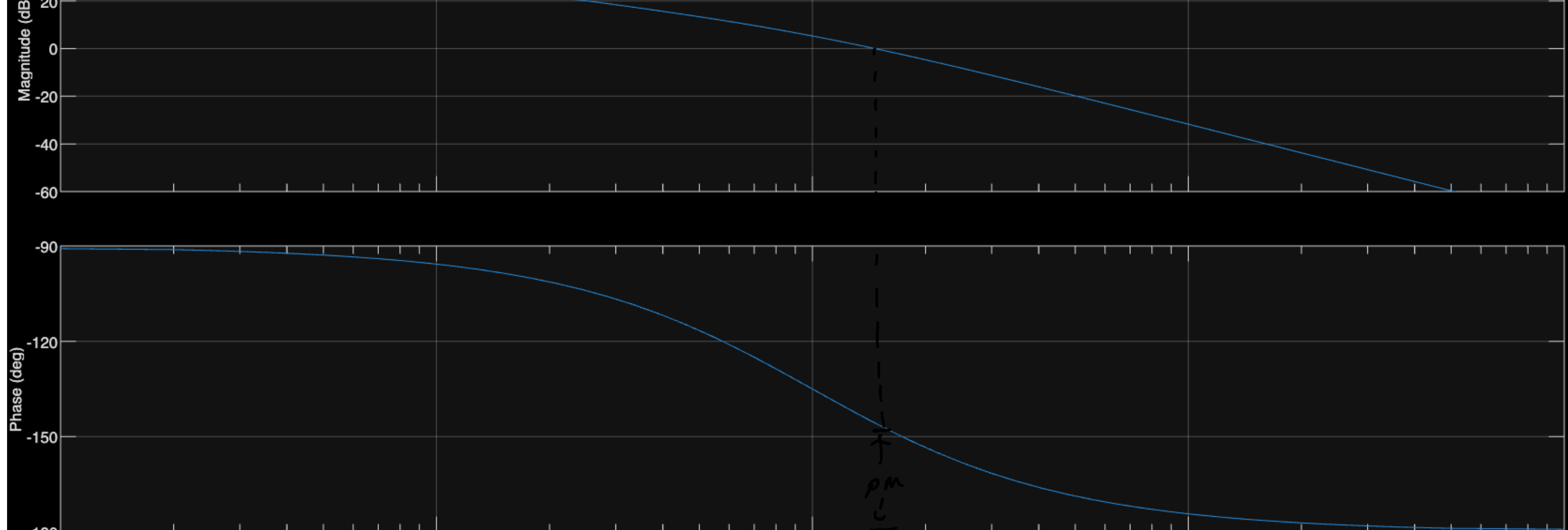
$|K G(j\omega)| = 1$

$1 = \frac{100K}{\sqrt{(100 - \omega^2)^2 + (2\omega)^2}}$

$\sqrt{(100 - \omega^2)^2 + (2\omega)^2} = 100K$

$K = 0.28$

2.



b.  $\omega_{\text{cross}} = 13.5 \text{ rad/s}$

$\text{PM} = 32^\circ$   $\text{GM} = \infty$  because  $\phi$  never reaches  $-180^\circ$

Using Margin:  $\text{PM} = 34^\circ$   $\omega_{\text{cross}} = 14.7 \text{ rad/s}$   $\phi_{\text{cross}} = 0$

c.  $e_{ss} = \frac{1}{100}$

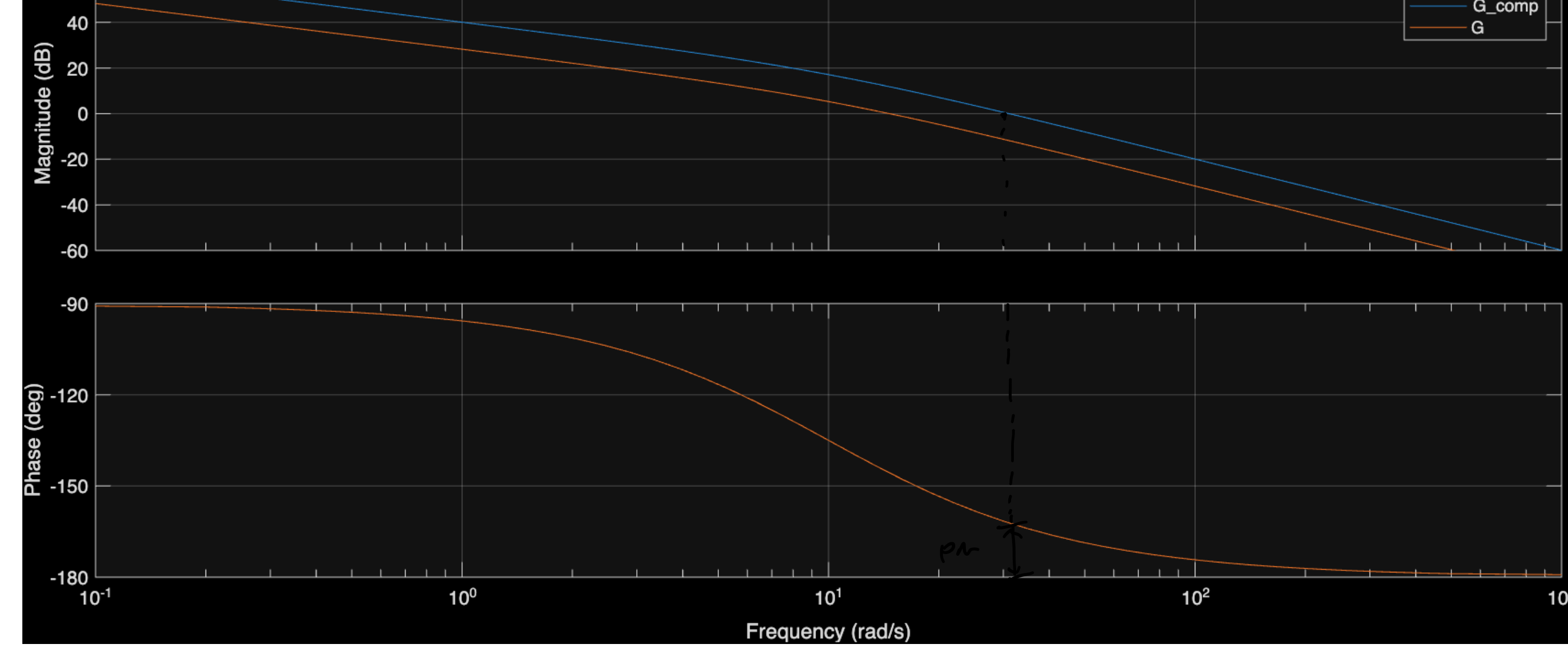
$K_a = 26$

$e_{ss} = \frac{1}{K_v}$  for step

$\frac{1}{K_v} < \frac{1}{100}$

$K_v \geq 100$

Total  $K_p = \frac{K_v}{K_a} = \frac{100}{26} = 3.86$



d.  $\omega_{\text{cross}} \approx 31 \text{ rad/s}$

$\omega_{\text{cross}} = 31.1 \text{ rad/s}$

$\text{PM} \approx 18^\circ$

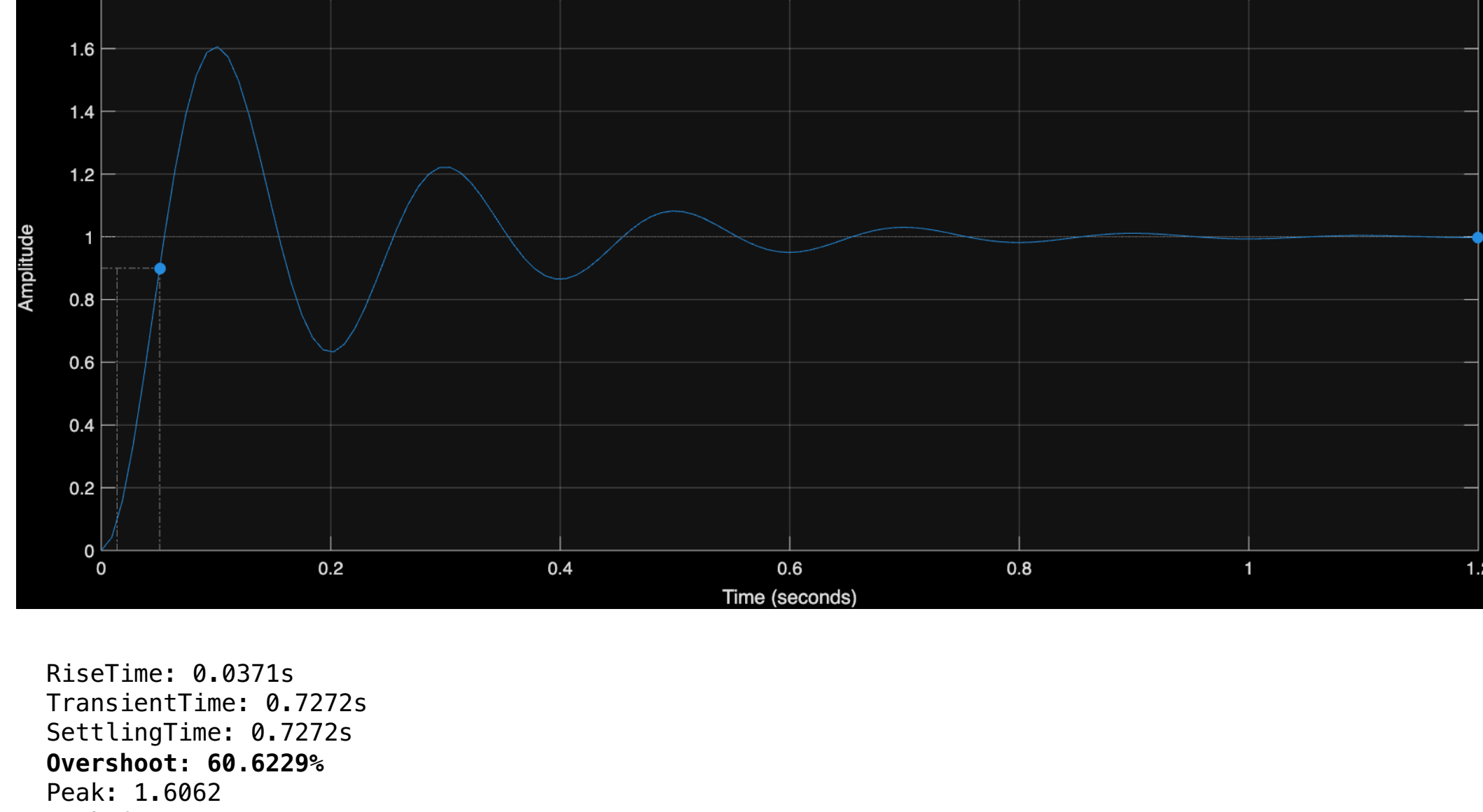
$\text{PM} = 17.8^\circ$

→ stable

by Inspection

by Margin () tool

e.



RiseTime: 0.0371s  
TransientTime: 0.7272s  
SettlingTime: 0.7272s  
Overshoot: 60.6229%  
Peak: 1.6062  
PeakTime: 0.1013s  
Steady-state error: 0

f.



$e_{ss} \ll \frac{1}{100}$

g.  $M_p \leq 20\%$   $\omega_{\text{cross}} < 14.7 \cdot 2 = 29.4 \text{ rad/s}$

$e_{ss} < \frac{1}{100}$  for ramp

determine  $P$  and  $z$   $|P| > |z|$

$M_p = e^{-\pi \zeta / \sqrt{1 - \zeta^2}}$

$\ln M_p = \frac{-\pi \zeta}{\sqrt{1 - \zeta^2}}$

$\ln M_p = \frac{-(\pi \zeta)^2}{1 - \zeta^2}$

$\zeta = \frac{\sqrt{(\ln(M_p))^2 / (\pi^2 + (\ln(M_p))^2)}}{\sqrt{1 + (\ln(M_p))^2 / \pi^2}} \rightarrow \text{required phase margin}$

$\sin \phi_{\text{max}} = \frac{1 - \zeta/p}{1 + \zeta/p}$  call  $\frac{\zeta}{p} = c$

$\sin \phi_{\text{max}} = \frac{1 - c}{1 + c}$   $p = \frac{\zeta}{c}$

$\sin \phi_{\text{max}} + c \sin \phi_{\text{max}} = 1 - c$

$c(1 + \sin \phi_{\text{max}}) = 1 - \sin \phi_{\text{max}}$

$\frac{\zeta}{p} = c = \frac{1 - \sin \phi_{\text{max}}}{1 + \sin \phi_{\text{max}}}$

at  $\omega_{\text{cross}} = 29.4 \text{ rad/s}$

$|G| = 0.2864$  and  $\phi_{cp} = -161.16^\circ$

$\text{PM}_{cp} = 180 + \phi_{cp}$

$\text{PM}_{\text{required}} < 180 + \phi_{cp} + \phi_{\text{max}}$

$\phi_{\text{max}} = \text{PM} - (180 + \phi_{cp})$

$\omega_{\text{max}} = \sqrt{\frac{z}{p}}$   $K_c = \frac{c}{|G| \sqrt{\alpha}}$  (looked up how to do this)

$z = \frac{\zeta}{\sqrt{\alpha}}$

$p = \frac{\zeta}{c}$

evaluate using matlab:

$K_c = 6.266$

$z = 16.3282$

$p = 52.6117$

$\omega_c = 29.4 \text{ rad/s}$  ✓

$e_{ss} = \frac{1}{K_c K_p 26} = 0.001596 < 0.01$  ✓

$M_p \% = 19.85\% < 20\%$  ✓

